

Jaspreet Singh Nahal

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

AI Developer — LLMs, RAG Systems, FastAPI — with internship experience and **350+ LeetCode problems**. Built conversational AI, trading systems, and automation tools using Python, React, FastAPI, and LangChain with focus on scalable architecture and performance optimization.

EXPERIENCE

Machine Learning Intern

Aug 2025 – Oct 2025

Expedien eSolutions

Noida, India

- Built RAG-based legal query resolver using LangChain, FAISS, and Hugging Face models with RetrievalQA pipeline
- Optimized document retrieval achieving <2s response time on 10K+ document corpus with Streamlit UI

Algorithmic Trading Developer

Nov 2024 – Dec 2024

Freelance

Remote

- Developed automated trading system with SMA crossover strategy using Flask API and PostgreSQL
- Reduced trade execution latency by 30% through async processing and optimized data handling

PROJECTS

EverydayGPT | *PyTorch, Transformers, FAISS, RAG, NLP*

Feb 2026

- Designed and pretrained a 205M-parameter GPT-style Transformer from scratch in PyTorch with causal self-attention, custom initialization, and GPT-2 tokenization on a 10B-token FineWeb-Edu + Wikipedia corpus.
- Built a Retrieval-Augmented Generation (RAG) pipeline using SentenceTransformers and FAISS, achieving sub-205ms retrieval latency and reducing hallucinations via confidence-based gating.
- Implemented instruction fine-tuning, selective loss masking, top-k sampling, and repetition control for stable multi-turn conversational performance.

CodeAtlas | *Python, FastAPI, NetworkX, React, TypeScript*

Apr 2026

- Built a full-stack code intelligence platform that ingests GitHub repos/ZIPs/code snippets and generates interactive multi-language dependency graphs.
- Implemented async FastAPI job processing and a filterable React force-graph UI to highlight architecture domains, key files, and dependency cycles in large repositories.

AI Assistant with NER | *React.js, Python, Flask, MongoDB*

Sep 2024

- Developed NLP conversational assistant with 92% intent recognition accuracy handling 1,000+ user sessions
- Integrated Named Entity Recognition and real-time Flask API with <1s response time

EDUCATION

B.Tech in Computer Science Engineering

2023 – 2027

Dr. A.P.J. Abdul Kalam Technical University, Uttar Pradesh

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript

ML & LLM Systems: PyTorch, Hugging Face Transformers, SentenceTransformers, FAISS, LangChain, Retrieval-Augmented Generation (RAG)

Data & Training: NumPy, Pandas, scikit-learn, Spark, gradient-based optimization, large-scale pretraining

Backend & Tooling: FastAPI, Flask, REST APIs, PostgreSQL, MongoDB, Git, Docker, GitHub Actions, Streamlit

HACKATHONS & CERTIFICATIONS

Competitive Achievement: 3rd Place – Voyager Blitz, HackIndia Spark-4 (HackerRank, Mar 2026)

Hackathons: IBM AI Hackathon (2025), Google Solutions Challenge (2025), GUVI Hackathon (2025)

Certifications: Google – Prompt Design in Vertex AI & Responsible AI; IBM – Accelerating Deep Learning with GPUs; BCG X (Forage) – GenAI Financial Chatbot Simulation